

Discrete Mathematics

A Concise 10-Week Course

Prerequisites: Calculus I–II, Linear Algebra.
Style: direct, no fluff — definitions, examples, practice.

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1 Week 1: Propositional Logic

1.1 Propositions and Connectives

A **proposition** is a statement that is either true (T) or false (F).

Core connectives

$\neg p$	negation	NOT p
$p \wedge q$	conjunction	p AND q
$p \vee q$	disjunction	p OR q
$p \rightarrow q$	implication	if p then q
$p \leftrightarrow q$	biconditional	p iff q

1.2 Truth Tables

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Key facts:

- $p \rightarrow q \equiv \neg p \vee q$ (implication as disjunction).
- $p \rightarrow q \equiv \neg q \rightarrow \neg p$ (contrapositive — always equivalent).
- $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$.

1.3 Logical Equivalences

Key equivalences

$\neg(\neg p) \equiv p$	Double negation
$p \wedge q \equiv q \wedge p, \quad p \vee q \equiv q \vee p$	Commutativity
$\neg(p \wedge q) \equiv \neg p \vee \neg q$	De Morgan 1
$\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan 2
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributivity
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	Distributivity
$p \wedge \mathbf{T} \equiv p, \quad p \vee \mathbf{F} \equiv p$	Identity
$p \wedge \mathbf{F} \equiv \mathbf{F}, \quad p \vee \mathbf{T} \equiv \mathbf{T}$	Domination

Example 1.1. Show $p \rightarrow q \equiv \neg p \vee q$.

Truth table: when $p=T, q=F$ — both give F; all other rows agree. ✓

1.4 Disjunctive Normal Form (DNF)

A formula is in **DNF** if it is a disjunction of conjunctions of literals (a literal is p or $\neg p$).

Algorithm (truth-table method):

```

1 Input:  formula F with variables p1,...,pn
2 Output: equivalent DNF
3
4 1. Build full truth table for F.
5 2. For each row where F = TRUE:
6   a. Form a minterm: conjoin each variable
7     as pi if its value is T, else neg(pi).
8 3. DNF = disjunction of all minterms.

```

Listing 1: DNF conversion algorithm

Example 1.2. Find DNF of $p \rightarrow q$. Truth table rows where result is T: (T,T) , (F,T) , (F,F) .

Minterms: $(p \wedge q)$, $(\neg p \wedge q)$, $(\neg p \wedge \neg q)$.

DNF: $(p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$.

Simplified (optional): $\neg p \vee q \equiv p \rightarrow q$. ✓

1.5 Applications

- **Logic circuits:** AND/OR/NOT gates implement $\wedge, \vee, \neg$.
- **Search queries:** “AI AND (NOT robot)” uses propositional logic.

Practice:

1. Build a truth table for $(p \rightarrow q) \wedge (\neg p \rightarrow r)$.
2. Prove $p \vee (p \wedge q) \equiv p$ (absorption) using equivalences.
3. Convert $\neg(p \rightarrow q)$ to DNF.
4. Show that $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$.
5. Is $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ a tautology?

2 Week 2: Sets and Relations

2.1 Set Operations

Core operations

$A \cup B$	union	all elements in A or B
$A \cap B$	intersection	elements in both
$A \setminus B$	difference	in A but not B
$\bar{A}$ (or A^c)	complement	$U \setminus A$
$\mathcal{P}(A)$	power set	all subsets of A

Key facts: $|\mathcal{P}(A)| = 2^{|A|}$. $A \cap \emptyset = \emptyset$. $A \cup \emptyset = A$.

De Morgan for sets: $\overline{A \cup B} = \bar{A} \cap \bar{B}$; $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

2.2 Binary Relations

Definition 2.1. A *binary relation* R on A is a subset $R \subseteq A \times A$. Write aRb iff $(a, b) \in R$.

Properties

Reflexive	$\forall a : aRa$
Symmetric	$aRb \Rightarrow bRa$
Antisymmetric	$aRb \wedge bRa \Rightarrow a = b$
Transitive	$aRb \wedge bRc \Rightarrow aRc$

Example 2.2. On $\mathbb{Z}$: $aRb \Leftrightarrow a \leq b$ is reflexive, antisymmetric, transitive (not symmetric).
 $aRb \Leftrightarrow a \equiv b \pmod{n}$ is reflexive, symmetric, transitive.

2.3 Equivalence Relations

An **equivalence relation** is reflexive + symmetric + transitive.

- The **equivalence class** of a : $[a] = \{b \in A \mid aRb\}$.
- Classes form a **partition** of A (disjoint, cover A).

Example 2.3. $\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}$ under $\equiv \pmod{n}$ partitions $\mathbb{Z}$ into n classes.

2.4 Partial Orders and Hasse Diagrams

A **partial order** is reflexive, antisymmetric, transitive. Notation: $(A, \preceq)$.

A **total order**: additionally $\forall a, b: a \preceq b$ or $b \preceq a$.

Hasse diagram: draw a below b if $a \prec b$ and no c satisfies $a \prec c \prec b$; omit self-loops and transitive edges.

Example 2.4. Divisibility on $\{1, 2, 3, 6\}$: $1 \mid 2, 1 \mid 3, 1 \mid 6, 2 \mid 6, 3 \mid 6$.

Hasse diagram has 6 at top, 2 and 3 in middle, 1 at bottom.

Practice:

1. Prove $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.
2. Is “ x is a sibling of y ” an equivalence relation on people?
3. List all equivalence classes of $\equiv \pmod{4}$ on $\{0, 1, \dots, 11\}$.

4. Draw the Hasse diagram for divisibility on $\{1, 2, 4, 8\}$.
5. How many relations on $\{1, 2, 3\}$ are both symmetric and reflexive?

3 Week 3: Functions and Cardinality

3.1 Types of Functions

Injective / Surjective / Bijective

Let $f : A \rightarrow B$.

- **Injective** (one-to-one): $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$.
- **Surjective** (onto): $\forall b \in B, \exists a \in A : f(a) = b$.
- **Bijective**: injective AND surjective.

Example 3.1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$: neither (not injective: $f(1) = f(-1)$; not surjective: -1 unreachable).

$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 1$: bijective.

$f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = n + 1$: bijective.

3.2 Inverse and Composition

- f^{-1} exists $\Leftrightarrow f$ is bijective.
- **Composition**: $(g \circ f)(a) = g(f(a))$.
- If $f : A \rightarrow B$ and $g : B \rightarrow C$ are both bijective, so is $g \circ f$.

3.3 Cardinality

Definition 3.2. $|A| = |B|$ if there exists a bijection $f : A \rightarrow B$.

Cardinality facts

- $|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}|$ — all **countably infinite** ($\aleph_0$).
- $|\mathbb{R}| > |\mathbb{N}|$ — $\mathbb{R}$ is **uncountable**.
- $\mathcal{P}(A)$ always has $|\mathcal{P}(A)| > |A|$ (Cantor's theorem).

Cantor's diagonal argument (sketch): Suppose $\mathbb{R}$ is countable; list all reals in $[0, 1]$ as $r_1, r_2, \dots$. Construct d with $d_i \neq (r_i)_i$ at each decimal digit. Then d differs from every r_i — contradiction. Hence $|\mathbb{R}| > \aleph_0$.

Example 3.3. $|\mathbb{N}| = |\mathbb{Z}|$: use bijection $f(n) = n/2$ if n even, $f(n) = -(n+1)/2$ if n odd.

Listing: $0, 1, -1, 2, -2, 3, -3, \dots$

Practice:

1. Is $f : \mathbb{N} \rightarrow \mathbb{N}, f(n) = n + 1$ injective? Surjective?
2. Let $f(x) = 3x - 1$ and $g(x) = x^2$. Find $(f \circ g)(2)$ and $(g \circ f)(2)$.
3. Prove $|\{0, 1\}^*| = |\mathbb{N}|$ (all binary strings are countable).
4. Show $|\mathcal{P}(\{1, 2, 3, 4\})| = 16$.
5. Let $f : A \rightarrow B$ be injective and $g : B \rightarrow C$ surjective. Is $g \circ f$ necessarily surjective?

4 Week 4: Integers and Number Theory

4.1 Divisibility and Primes

Definition 4.1. $a \mid b$ (“ a divides b ”) iff $\exists k \in \mathbb{Z} : b = ka$.

Theorem 4.2 (Infinitely many primes — Euclid). *Suppose finitely many primes $p_1, \dots, p_k$. Let $N = p_1 \cdots p_k + 1$. No p_i divides N (remainder 1), so N has a prime factor not in the list — contradiction.*

4.2 Euclidean Algorithm

$\gcd(a, b) = \gcd(b, a \bmod b)$; base case $\gcd(a, 0) = a$.

```

1 gcd(a, b):
2   while b != 0:
3     a, b = b, a mod b
4   return a    // gcd
5
6 extended_gcd(a, b): // returns (g, x, y) s.t. ax+by=g
7   if b == 0: return (a, 1, 0)
8   (g, x1, y1) = extended_gcd(b, a mod b)
9   return (g, y1, x1 - (a div b)*y1)

```

Listing 2: Euclidean / Extended Euclidean algorithm

Example 4.3. $\gcd(48, 18)$: $48 = 2 \cdot 18 + 12$; $18 = 1 \cdot 12 + 6$; $12 = 2 \cdot 6 + 0$. So $\gcd = 6$.
Extended: $6 = 18 - 1 \cdot 12 = 18 - 1 \cdot (48 - 2 \cdot 18) = 3 \cdot 18 - 1 \cdot 48$. So $x = -1, y = 3$.

4.3 Modular Arithmetic

$a \equiv b \pmod{m} \Leftrightarrow m \mid (a - b)$.

Arithmetic mod m

$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$; same for $\times$.

Modular inverse of a : a^{-1} s.t. $a \cdot a^{-1} \equiv 1 \pmod{m}$; exists iff $\gcd(a, m) = 1$.

4.4 Linear Congruences

$ax \equiv b \pmod{m}$ has a solution iff $d = \gcd(a, m) \mid b$.

If so, there are exactly d solutions mod m . Divide through by d ; use extended GCD.

Example 4.4. $6x \equiv 4 \pmod{14}$. $d = \gcd(6, 14) = 2$, $2 \mid 4$. Divide: $3x \equiv 2 \pmod{7}$.
 $3^{-1} \pmod{7}$: $3 \cdot 5 = 15 \equiv 1$, so $x \equiv 5 \cdot 2 = 10 \equiv 3 \pmod{7}$.
Solutions mod 14: $x = 3$ and $x = 3 + 7 = 10$.

4.5 Chinese Remainder Theorem (CRT)

Theorem 4.5 (CRT — two moduli). *If $\gcd(m_1, m_2) = 1$, the system $x \equiv a_1 \pmod{m_1}$, $x \equiv a_2 \pmod{m_2}$ has a unique solution mod $m_1 m_2$.*

Construction: $M = m_1 m_2$, $M_1 = m_2$, $M_2 = m_1$. Find $y_i = M_i^{-1} \pmod{m_i}$.
 Solution: $x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 \pmod{M}$.

Example 4.6. $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$. $M = 15$, $M_1 = 5$, $M_2 = 3$.

$$5y_1 \equiv 1 \pmod{3} \Rightarrow 2y_1 \equiv 1 \Rightarrow y_1 = 2.$$

$$3y_2 \equiv 1 \pmod{5} \Rightarrow y_2 = 2.$$

$$x \equiv 2 \cdot 5 \cdot 2 + 3 \cdot 3 \cdot 2 = 20 + 18 = 38 \equiv 8 \pmod{15}.$$

4.6 Linear Diophantine Equations

$ax + by = c$ has integer solutions iff $\gcd(a, b) \mid c$. Find one solution via extended GCD; general solution: $x = x_0 + \frac{b}{d}t$, $y = y_0 - \frac{a}{d}t$, $t \in \mathbb{Z}$.

Practice:

1. Compute $\gcd(252, 105)$ using the Euclidean algorithm.
2. Solve $7x \equiv 1 \pmod{11}$ (find modular inverse).
3. Solve $15x \equiv 9 \pmod{25}$.
4. Solve the system: $x \equiv 3 \pmod{4}$, $x \equiv 5 \pmod{7}$ using CRT.
5. Solve the Diophantine equation $12x + 9y = 21$.

5 Week 5: Induction and Recursion

5.1 Mathematical Induction Templates

Ordinary induction

To prove $P(n)$ for all $n \geq n_0$:

1. **Base:** prove $P(n_0)$.
2. **Step:** assume $P(k)$ (inductive hypothesis); prove $P(k+1)$.

Strong induction

Step: assume $P(n_0), P(n_0+1), \dots, P(k)$; prove $P(k+1)$.

Use when $P(k+1)$ depends on earlier terms, not just $P(k)$.

Example 5.1. Prove $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

Base ($n=1$): $1 = 1(2)/2$. Step: assume true for k . Then $\sum_{i=1}^{k+1} i = \frac{k(k+1)}{2} + (k+1) = (k+1)\frac{k+2}{2}$.

5.2 Recursive Sequences

- Fibonacci: $F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$.
- Factorial: $0! = 1, n! = n \cdot (n-1)!$.

5.3 Linear Recurrences — Characteristic Polynomial

For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ (order 2, constant coefficients), find roots of $r^2 - c_1 r - c_2 = 0$.

Three cases

- I. **Distinct real roots** $r_1 \neq r_2$: $a_n = Ar_1^n + Br_2^n$.
- II. **Repeated root** $r_1 = r_2 = r$: $a_n = (A + Bn)r^n$.
- III. **Complex roots** $r = \alpha \pm \beta i$ (briefly): $a_n = \rho^n (A \cos n\theta + B \sin n\theta)$ where $\rho = |r|, \theta = \arg r$.
(Focus is on cases I–II.)

Use initial conditions a_0, a_1 to find A, B .

Example 5.2. Fibonacci: $a_n = a_{n-1} + a_{n-2}$, so $r^2 - r - 1 = 0 \Rightarrow r = \frac{1 \pm \sqrt{5}}{2}$.

$\varphi = \frac{1+\sqrt{5}}{2}, \hat{\varphi} = \frac{1-\sqrt{5}}{2}$. General: $F_n = A\varphi^n + B\hat{\varphi}^n$. Using $F_0 = 0, F_1 = 1$: $A = 1/\sqrt{5}, B = -1/\sqrt{5}$.

$$F_n = \frac{\varphi^n - \hat{\varphi}^n}{\sqrt{5}} \quad (\text{Binet's formula}).$$

Example 5.3. Solve $a_n = 4a_{n-1} - 4a_{n-2}$, $a_0 = 1, a_1 = 4$.

$r^2 - 4r + 4 = 0 \Rightarrow (r-2)^2 = 0 \Rightarrow r = 2$ (repeated).

$a_n = (A + Bn)2^n$. From $a_0 = A = 1$; $a_1 = 2(A + B) = 4 \Rightarrow B = 1$.

$a_n = (1+n)2^n$.

Practice:

1. Prove by induction: $\sum_{i=0}^n 2^i = 2^{n+1} - 1$.
2. Prove by strong induction: every $n \geq 2$ has a prime factor.
3. Solve the recurrence $a_n = 5a_{n-1} - 6a_{n-2}$, $a_0 = 1, a_1 = 0$.

4. Solve $a_n = 6a_{n-1} - 9a_{n-2}$, $a_0 = 0, a_1 = 3$.
5. Find a closed form for $a_n = a_{n-1} + 2$, $a_0 = 5$.

6 Week 6: Counting Principles

6.1 Sum and Product Rules

Sum and Product Rules

Sum: if task can be done in m ways or n ways (disjoint alternatives), total = $m + n$.

Product: if task has k sequential steps with $n_1, n_2, \dots, n_k$ choices each, total = $n_1 \cdot n_2 \cdots n_k$.

Example 6.1. Passwords: 6 characters, each uppercase letter or digit. $36^6 = 2,176,782,336$ passwords (product rule).

License plates: 3 letters then 3 digits: $26^3 \cdot 10^3 = 17,576,000$ (product rule).

6.2 Inclusion–Exclusion Principle

Inclusion–Exclusion

Two sets: $|A \cup B| = |A| + |B| - |A \cap B|$.

Three sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Example 6.2. Students who study French or German: $|F| = 25, |G| = 30, |F \cap G| = 10$.
 $|F \cup G| = 25 + 30 - 10 = 45$.

Example 6.3. Integers 1–100 divisible by 2, 3, or 5:

$|A| = 50, |B| = 33, |C| = 20; |A \cap B| = 16, |A \cap C| = 10, |B \cap C| = 6; |A \cap B \cap C| = 3$.

$|A \cup B \cup C| = 50 + 33 + 20 - 16 - 10 - 6 + 3 = 74$.

6.3 Pigeonhole Principle

Pigeonhole

Simple: if $n + 1$ objects go into n boxes, some box has ≥ 2 objects.

Generalized: if N objects go into k boxes, some box has $\geq \lceil N/k \rceil$ objects.

Example 6.4. Among any 367 people, at least 2 share a birthday (365 possible days, $\lceil 367/365 \rceil = 2$).

Example 6.5. Among 100 integers, at least $\lceil 100/10 \rceil = 10$ have the same last digit (10 possible last digits).

Example 6.6. In a group of 5 points inside a 2×2 square, by pigeonhole (4 unit squares), at least 2 points are within distance $\sqrt{2}$ of each other.

Practice:

1. How many bit strings of length 8 start with 1 or end with 00?
2. In a city of 3 million people, must at least two people have the same number of hairs? (Hair count $< 200,000$.)
3. 100 integers are chosen from $\{1, \dots, 200\}$. Prove two are consecutive.
4. Students: 120 study math, 80 study CS, 40 study both. How many study math or CS?
5. How many integers 1–1000 are divisible by 3 or 5 but not by 15?

7 Week 7: Permutations and Combinations

7.1 Permutations

Permutations $P(n, k)$

Ordered selections of k items from n distinct items:

$$P(n, k) = \frac{n!}{(n-k)!} = n(n-1) \cdots (n-k+1).$$

Example 7.1. Ways to arrange 3 books from a shelf of 7: $P(7, 3) = 7 \cdot 6 \cdot 5 = 210$.

7.2 Combinations

Combinations $C(n, k)$

Unordered selections of k items from n distinct items:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{P(n, k)}{k!}.$$

Example 7.2. Committee of 3 from 7 people: $\binom{7}{3} = 35$. $\binom{n}{0} = \binom{n}{n} = 1$.

7.3 Combinations with Repetition

Choosing k items from n types, repetition allowed, order doesn't matter:

$$C(n+k-1, k) = \binom{n+k-1}{k}.$$

Example 7.3. Buy 6 donuts from 4 flavors: $\binom{4+6-1}{6} = \binom{9}{6} = 84$.

7.4 Binomial Theorem

Binomial Theorem

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Example 7.4. $(x+y)^4 = \binom{4}{0}y^4 + \binom{4}{1}xy^3 + \binom{4}{2}x^2y^2 + \binom{4}{3}x^3y + \binom{4}{4}x^4 = y^4 + 4xy^3 + 6x^2y^2 + 4x^3y + x^4$.

7.5 Pascal's Identity

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Combinatorial proof: either element n is in the subset (choose $k-1$ from $n-1$) or it is not (choose k from $n-1$).

Useful identities:

- $\sum_{k=0}^n \binom{n}{k} = 2^n$ (set $x = y = 1$).
- $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$ (set $x = -1, y = 1$).

Practice:

1. How many 5-card hands from a 52-card deck?
2. Prove $\binom{n}{k} = \binom{n}{n-k}$.
3. How many ways to distribute 10 identical balls into 4 distinct boxes?
4. Find the coefficient of x^3y^5 in $(x+y)^8$.
5. Find the coefficient of x^4 in $(1+x)^{10}$.

8 Week 8: Graph Theory — Definitions and Isomorphism

8.1 Basic Definitions

Graph vocabulary

Simple graph	$G = (V, E)$: no loops, no multi-edges
Multigraph	allows multiple edges between same pair
Degree $\deg(v)$	number of edges incident to v
$\delta(G)$	minimum degree; $\Delta(G)$ maximum degree
Complete K_n	every pair of vertices connected
Bipartite	$V = X \cup Y$, edges only between X and Y

Theorem 8.1 (Handshaking Lemma). $\sum_{v \in V} \deg(v) = 2|E|$.

Corollary: the number of odd-degree vertices is even.

Example 8.2. K_4 : 4 vertices, $\binom{4}{2} = 6$ edges; each vertex has degree 3. Sum of degrees: $4 \cdot 3 = 12 = 2 \cdot 6$.

8.2 Graph Isomorphism

Definition 8.3. $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are **isomorphic** if there exists a bijection $f : V_1 \rightarrow V_2$ such that $\{u, v\} \in E_1 \Leftrightarrow \{f(u), f(v)\} \in E_2$.

Necessary conditions (not sufficient!):

- Same number of vertices and edges.
- Same degree sequence (sorted).
- Same number of cycles, paths, connected components.

Example 8.4. Two graphs on 4 vertices: G_1 is a 4-cycle (C_4 , degree sequence $[2, 2, 2, 2]$) and G_2 is a path P_4 (degree sequence $[1, 2, 2, 1]$). Different degree sequences $\Rightarrow$ not isomorphic.

Example 8.5. G_1 : vertices $\{a, b, c, d\}$, edges ab, bc, cd, da . G_2 : vertices $\{1, 2, 3, 4\}$, edges $12, 23, 34, 41$. Both C_4 ; map $a \mapsto 1, b \mapsto 2, c \mapsto 3, d \mapsto 4$. Isomorphic.

Checklist for isomorphism:

1. Check $|V|, |E|$, degree sequences.
2. Check cycles of each length.
3. Try to construct explicit bijection.

Practice:

1. Prove K_5 has 10 edges and each vertex has degree 4.
2. A graph has 6 vertices, degrees 2, 2, 3, 3, 3, 3. How many edges?
3. Are the following isomorphic? G_1 : C_6 (6-cycle). G_2 : two triangles sharing no vertices.
4. How many non-isomorphic simple graphs are there on 3 vertices?
5. Show that if a graph has exactly two vertices of odd degree, those vertices are connected by a path.

9 Week 9: Paths and Circuits

9.1 Terminology

Walk / Trail / Path / Circuit	
Walk	sequence of vertices, edges can repeat
Trail	walk with no repeated edge
Path	walk with no repeated vertex
Circuit	closed trail (starts = ends, no repeated edge)
Cycle	closed path (no repeated vertex except start=end)

9.2 Eulerian Circuits

Theorem 9.1 (Euler's Theorem). *A connected graph has an **Eulerian circuit** (traverses every edge exactly once, returns to start) if and only if every vertex has even degree.*

*It has an **Eulerian trail** (doesn't need to return) iff it has exactly 0 or 2 vertices of odd degree.*

Example 9.2. *The Königsberg Bridge problem: 4 vertices, degrees 3, 3, 5, 3 (all odd). No Eulerian trail exists.*

K_4 : all degrees 3 (odd). No Eulerian circuit; 2 odd-degree vertices needed for trail — fails.

9.3 Hamiltonian Cycles

Definition 9.3. *A **Hamiltonian cycle** visits every vertex exactly once (returns to start).*

No simple characterization like Euler's exists. A sufficient condition:

Theorem 9.4 (Dirac's Theorem). *If G is simple, $|V| \geq 3$, and every vertex satisfies $\deg(v) \geq |V|/2$, then G has a Hamiltonian cycle.*

Dirac's condition is *sufficient* but not *necessary*. C_5 is Hamiltonian but $\delta = 2 < 5/2 = 2.5$.

9.4 Adjacency Matrix and Walks

The **adjacency matrix** A of G has $A_{ij} = 1$ if $\{i, j\} \in E$, else 0.

Powers of adjacency matrix

$(A^k)_{ij}$ = number of walks of length k from i to j .

Example 9.5. C_4 (4-cycle 1–2–3–4–1):

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \end{pmatrix}.$$

$(A^2)_{13} = 2$: walks 1–2–3 and 1–4–3.

9.5 Euler vs. Hamiltonian

	Eulerian	Hamiltonian
Visits	every <i>edge</i> once	every <i>vertex</i> once
Characterization	clean (degree parity)	no clean iff (NP-hard)
Algorithm	Hierholzer's $O(E)$	no poly-time known in general

Practice:

1. Does K_5 have an Eulerian circuit? An Eulerian trail?
2. Does $K_{3,3}$ (complete bipartite) have a Hamiltonian cycle?
3. Compute A^2 for the path P_3 (1–2–3) and interpret $(A^2)_{13}$.
4. For which n does K_n have an Eulerian circuit?
5. Verify Dirac's condition for K_5 and conclude it is Hamiltonian.

10 Week 10: Planar Graphs and Graph Coloring

10.1 Planar Graphs

Definition 10.1. G is **planar** if it can be drawn in the plane with no edge crossings.

Theorem 10.2 (Euler's Formula). For a connected planar graph: $V - E + F = 2$, where F = number of faces (including the outer face).

Example 10.3. K_4 : $V = 4, E = 6$. Drawn planar: $F = 4$. $4 - 6 + 4 = 2$.

Corollary: For a simple planar graph with $V \geq 3$: $E \leq 3V - 6$.

For bipartite planar graphs with $V \geq 3$: $E \leq 2V - 4$.

Theorem 10.4 (Kuratowski's Theorem). G is non-planar iff it contains a subdivision of K_5 or $K_{3,3}$ as a subgraph.

Example 10.5. K_5 : $V = 5, E = 10$. $3V - 6 = 9 < 10$, so K_5 is non-planar by the corollary.
 $K_{3,3}$: $V = 6, E = 9$. $2V - 4 = 8 < 9$ (bipartite bound), so $K_{3,3}$ is non-planar.

10.2 Graph Coloring

Definition 10.6. A **proper k -coloring** assigns colors $\{1, \dots, k\}$ to vertices so adjacent vertices get different colors. The **chromatic number** $\chi(G)$ is the minimum k .

Key chromatic numbers

- $\chi(K_n) = n$: all pairs adjacent, all different colors.
- $\chi(\text{bipartite}) = 2$ (if at least one edge; 1 if no edges).
- $\chi(C_n) = 2$ if n even; $= 3$ if n odd.
- **4-Color Theorem:** every planar graph satisfies $\chi(G) \leq 4$.

10.3 Greedy Coloring Algorithm

```

1 Input:  G = (V, E), vertex ordering v1, v2, ..., vn
2 Output: coloring c[v] for each vertex
3
4 for i = 1 to n:
5     forbidden = {c[u] : u in neighbors(vi) and u already colored}
6     c[vi] = smallest positive integer not in forbidden
7 return c

```

Listing 3: Greedy graph coloring

Properties:

- Uses at most $\Delta(G) + 1$ colors.
- Result depends on vertex ordering; not always optimal.
- Running time: $O(|V| + |E|)$.

Example 10.7. C_5 (5-cycle, $v_1-v_5-v_4-v_3-v_2-v_1$): Order v_1, v_2, v_3, v_4, v_5 .
 $c(v_1) = 1$; $c(v_2) = 2$ (v_1 forbidden); $c(v_3) = 1$ (v_2 forbidden);
 $c(v_4) = 2$ (v_3 forbidden); $c(v_5) = 3$ ($v_1 = 1$ and $v_4 = 2$ forbidden).
 Greedy uses 3 colors. Optimal: $\chi(C_5) = 3$.

Practice:

1. Verify Euler's formula for the octahedron ($V = 6, E = 12$).
2. Is $K_{3,3}$ planar? Use the bipartite corollary to show not.
3. Find $\chi(K_{2,3})$ and $\chi(C_6)$.
4. Apply greedy coloring to K_4 in any vertex order.
5. A map has 5 regions where each pair of adjacent regions shares a border. What is the minimum number of colors to color it?

Solutions to Practice Problems

Week 1 — Propositional Logic

1. Build truth table with 8 rows; formula is not a tautology (F when $p=F, q=F, r=F$).
2. $p \vee (p \wedge q) \equiv (p \vee p) \wedge (p \vee q) \equiv p \wedge (p \vee q) \equiv p$. Use distributivity then absorption.
3. $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv p \wedge \neg q$. DNF: $p \wedge \neg q$ (single minterm).
4. Build truth table: both sides are T exactly when p, q have same value.
5. Yes — it is the **hypothetical syllogism** tautology. Check: when $p=T, q=F, r=T$, we get $((T \rightarrow F) \wedge (F \rightarrow T)) \rightarrow (T \rightarrow T) = (F \wedge T) \rightarrow T = F \rightarrow T = T$. All rows are T.

Week 2 — Sets and Relations

1. Let $x \in A \setminus (B \cap C)$. Then $x \in A$ and $x \notin B \cap C$, so $x \notin B$ or $x \notin C$. Hence $x \in (A \setminus B) \cup (A \setminus C)$. Reverse inclusion similar.
2. Not an equivalence relation: not reflexive (no one is their own sibling), not transitive in some definitions.
3. Classes: $\{0, 4, 8\}, \{1, 5, 9\}, \{2, 6, 10\}, \{3, 7, 11\}$.
4. Hasse: 8 on top; 4 below; then 2; then 1 at bottom. Single chain.
5. Reflexive + symmetric subsets of $\{1, 2, 3\}^2$: 3 reflexive pairs fixed; choose from 3 symmetric off-diagonal pairs $(\{12, 21\}, \{13, 31\}, \{23, 32\})$. Total: $2^3 = 8$ relations.

Week 3 — Functions

1. Injective (yes: $f(a) = f(b) \Rightarrow a + 1 = b + 1$). Not surjective (0 has no preimage in $\mathbb{N}$ with $\mathbb{N} = \{1, 2, \dots\}$; or surjective if $\mathbb{N} = \{0, 1, \dots\}$).
2. $(f \circ g)(2) = f(4) = 11$. $(g \circ f)(2) = g(5) = 25$.
3. Enumerate all finite binary strings by length, then lexicographically: $\epsilon, 0, 1, 00, 01, 10, 11, \dots$
4. $|\mathcal{P}(\{1, 2, 3, 4\})| = 2^4 = 16$.
5. No. Counterexample: $A = \{1\}, B = \{1, 2\}, C = \{1, 2\}$; $f(1) = 1$ (injective); $g(1) = 1, g(2) = 2$ (surjective); $(g \circ f)(1) = 1$ but $2 \in C$ not hit. So no.

Week 4 — Integers

1. $252 = 2 \cdot 105 + 42$; $105 = 2 \cdot 42 + 21$; $42 = 2 \cdot 21 + 0$. gcd = 21.
2. $7x \equiv 1 \pmod{11}$: $7 \cdot 8 = 56 = 5 \cdot 11 + 1 \equiv 1$. Inverse = 8.
3. $\gcd(15, 25) = 5$, $5 \mid 9$? No! So **no solution**.
4. $M = 28$; $M_1 = 7, y_1$: $7y_1 \equiv 1 \pmod{4} \Rightarrow 3y_1 \equiv 1 \Rightarrow y_1 = 3$. $M_2 = 4, y_2$: $4y_2 \equiv 1 \pmod{7} \Rightarrow y_2 = 2$. $x \equiv 3 \cdot 7 \cdot 3 + 5 \cdot 4 \cdot 2 = 63 + 40 = 103 \equiv 103 - 3 \cdot 28 = 103 - 84 = 19 \pmod{28}$.
5. $\gcd(12, 9) = 3$, $3 \mid 21$. $12x + 9y = 21 \Rightarrow 4x + 3y = 7$. One solution: $x = 4, y = -3$. General: $x = 4 + 3t, y = -3 - 4t$.

Week 5 — Induction and Recursion

1. Base $n = 0$: $1 = 2^1 - 1$. Step: $\sum_{i=0}^{k+1} 2^i = 2^{k+1} - 1 + 2^{k+1} = 2^{k+2} - 1$.
2. Base $n = 2$: 2 is prime. Step: let $n > 2$. If n prime, done. Otherwise $n = ab$ with $1 < a < n$; by strong hypothesis a has a prime factor, which divides n .
3. Char poly: $r^2 - 5r + 6 = 0 \Rightarrow r = 2, 3$. $a_n = A \cdot 2^n + B \cdot 3^n$. $a_0 = A + B = 1$; $a_1 = 2A + 3B = 0$. $B = -2, A = 3$. $a_n = 3 \cdot 2^n - 2 \cdot 3^n$.

4. Char poly: $(r - 3)^2 = 0 \Rightarrow r = 3$. $a_n = (A + Bn)3^n$. $a_0 = A = 0$; $a_1 = 3B = 3 \Rightarrow B = 1$.
 $a_n = n \cdot 3^n$.
5. $a_n = 5 + 2n$ (arithmetic sequence with $d = 2$).

Week 6 — Counting Principles

1. Start with 1: $2^7 = 128$. End with 00: $2^6 = 64$. Both: start 1, end 00 $\Rightarrow 2^5 = 32$. Total:
 $128 + 64 - 32 = 160$.
2. Yes: $3,000,000 > 200,000$, so by pigeonhole at least $\lceil 3,000,000/200,000 \rceil = 15$ people share
the same hair count.
3. Pigeonhole: 100 integers from 200; by PHP among any 100 consecutive pairs $\{(1, 2), (3, 4), \dots, (199, 200)\}$
— actually: partition into 100 pairs. Each chosen integer lands in one pair; with 100 integers
and 100 pairs, some pair is fully covered.
4. $|M \cup C| = 120 + 80 - 40 = 160$.
5. By 3: $\lfloor 1000/3 \rfloor = 333$. By 5: 200. By 15: 66. Div by 3 or 5: $333 + 200 - 66 = 467$. Div by
15: 66. Div by 3 or 5 but not 15: $467 - 66 = 401$.

Week 7 — Permutations and Combinations

1. $\binom{52}{5} = 2,598,960$.
2. $\binom{n}{n-k} = \frac{n!}{(n-k)!k!} = \binom{n}{k}$.
3. Stars and bars: $\binom{10+4-1}{10} = \binom{13}{10} = 286$.
4. $\binom{8}{3} = 56$.
5. $\binom{10}{4} = 210$.

Week 8 — Graph Theory

1. $E = \binom{5}{2} = 10$. Each vertex connects to 4 others. $\sum \deg = 5 \cdot 4 = 20 = 2 \cdot 10$.
2. $\sum \deg = 2 + 2 + 3 + 3 + 3 + 3 = 16$, so $|E| = 8$.
3. $G_1 = C_6$: degree sequence $[2, 2, 2, 2, 2, 2]$, connected, one component. G_2 : two triangles,
degree sequence $[2, 2, 2, 2, 2, 2]$, **two** components. Not isomorphic (different connectivity).
4. Non-isomorphic graphs on 3 vertices: 4 (0 edges, 1 edge, 2 edges (P_3), 3 edges (K_3)).
5. Use Euler's theorem on connected components or direct argument via induction.

Week 9 — Paths and Circuits

1. K_5 : all 5 degrees are 4 (even). Connected. Eulerian circuit exists. For Eulerian trail: 0 odd
vertices, so also has circuits (which are trails).
2. $K_{3,3}$: bipartite, 6 vertices, each degree 3 (odd). Eulerian circuit: no (odd degrees). Hamilto-
nian cycle: yes — can be verified directly (e.g., $1-a-2-b-3-c-1$).
3. $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$; $A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$. $(A^2)_{13} = 1$: one walk $1-2-3$.
4. K_n has all vertices of degree $n - 1$. Eulerian circuit requires all even degrees: $n - 1$ even iff
 n is odd. So K_n has Eulerian circuit iff n is odd.
5. K_5 : $n = 5$, $\delta = 4 \geq 5/2 = 2.5$. Dirac satisfied $\Rightarrow$ Hamiltonian cycle exists.

Week 10 — Planar Graphs and Coloring

1. Octahedron: $V = 6, E = 12, F = 8$. $6 - 12 + 8 = 2$.
2. $K_{3,3}$: $V = 6, E = 9$. Bipartite bound: $E \leq 2V - 4 = 8$. But $9 > 8$. Non-planar.
3. $\chi(K_{2,3}) = 2$ (bipartite with edges). $\chi(C_6) = 2$ (even cycle).
4. K_4 , order v_1, v_2, v_3, v_4 : $c(v_1) = 1, c(v_2) = 2, c(v_3) = 3, c(v_4) = 4$. Uses 4 colors = $\chi(K_4)$.
5. If each pair of 5 regions shares a border, the underlying graph is K_5 , which is non-planar. In planar maps, K_5 cannot occur, but the 4-color theorem still applies: 4 colors suffice. If the graph truly requires all pairs adjacent, $\chi = 5$; but such a configuration is non-planar and thus cannot arise from a planar map, so in practice 4 colors always suffice.